

First name:

Last name:

Student ID:

Name and Student ID must be filled in for this page to be graded.

Problem 1. Find $\text{curl}(\vec{v})$ with respect to right-handed coordinates x, y, z for

$$\vec{v} = \begin{bmatrix} 2y^2 \\ 5x \\ 0 \end{bmatrix}.$$

[5 marks]

Solution.

□

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Problem 2. Find a line integral of the form $\int_a^b f(\vec{r}(t)) dt$ where

$$f = xyz, \quad C : \vec{r} = [4t, 3t^2, 12t], \quad -2 \leq t \leq 2.$$

[5 marks]

Solution.

□

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Problem 3. Compute a line integral of the form $\int \vec{F} \cdot d\vec{r}$, where

$$\vec{F} = [xz, yz, xy], \quad C : \vec{r} = [t, t, e^t], \quad 0 \leq t \leq 5.$$

Hint: Integration by parts formula $\int u \, dv = uv - \int v \, du$.

[5 marks]

Solution.

□

MATH73235

Spring 2025

Quiz 3

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Extra Page: Must submit even if unused.